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PLC Programming Instructions

Programmable Logic Controllers

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Agenda

PLC Data Types

Basic PLC Programming Instructions

PLC Timers and Counters

Advanced PLC Programming Instructions

Data Manupulation Instructions

Student Learning Outcomes

1. Create programs in Ladder Logic using the following instructions in an application
2. Program PLC timers and counters

PLC Data Types

BIT

- 1 bit
- The smallest unit of memory
- Represents discrete binary states
  - 0 (OFF/FALSE)
  - 1 (ON/TRUE)

NIBBLE

- 4 bits
- 1 binary coded decimal (BCD) digit

BYTE

- 8 Bits
- 256 unique combinations
- Typically used for ASCII characters–1 BYTE = 1 character

WORD

- 16 Bits
- Historical standard register size in PLCs (AB SLC 500/Siemens S7-300)

DWORD/LWORD

- DWORD–32 bits
- LWORD–64 bits
- Modern architecture
- Used for high precision motion/math in older systems

Sub-Bit Access

- These data types are most often used as a number
- But they are *also* just a collection of bits
- We can access the individual bit inside a word using *dot notation*
  - `Word_Var.0`, `Word_Var.7`, etc.

IEC 61131-3 Elementary Types

Type	Keyword	Bits	Signed Range	Common Industrial Usage
Boolean	`BOOL`	1	0 or 1 (FALSE / TRUE)	Proximity sensors, pushbuttons, coils
Short Integer	`SINT`	8	-128 to +127	Small part counts, status flags
Integer	`INT`	16	-32,768 to +32,767	Analog inputs (0-10V, 4-20mA raw counts)
Double Integer	`DINT`	32	-2,147,483,648 to +2,147,483,647	High-speed encoders, totalizer counts
Real (Float)	`REAL`	32	IEEE 754 Floating Point	Scaled engineering units (PSI, GPM, °C)

Signed Integers

- Integers can be signed (+/-) or unsigned
- The defautl in most PLCs is signed–unsigned is optional
- In a signed integer the first bit is the sign: 0 = +, 1 = -
- Because we lose a bit, the capacity of the number is effectively cut in half
- `INT`: -32,768 to 32,767
- `UINT`: 0 to 65,535

Floating Point Numbers

- IEEE 754 Format
- 32-bits: 1 sign, 8 exponent bits, 23 mantissa bits
- An integer is *only* whole numbers, `REAL` must be used if a decimal point is necessary
- *NEVER* use an equality check (`REAL_1 = 10.0`) on a `REAL`–floating point numbers have in-built rounding errors

Addressing Memory in PLCs

- **Direct Memory Addressing:** Logic references explicit *physical* register locations in the CPU
- **Tag-Based Addressing:** Programmer creates descriptive, symbolic names stored in a database
- IEC 61131-3 standardizes both approaches
- Older PLCs pretty exclusively direct addressed, modern PLCs mostly use tag-based databases

Direct vs. Tag-Based Syntax

AB Direct (SLC/Micrologix)
Uses fixed file letters & numbers
`I:1/0`–Input slot 1, bit 0
`B3:0/0`–Binary bit word 0, bit 3
`N7:0`–Integer file 7, register 0

IEC 61131-3 Direct Syntax
Uses standard prefix notation
`%IX0.0`–Input byte 0, bit 0
`%QX0.0`–Output byte 0, bit 0
`%MW100`–Memory word 100

Tag-Based (Logix/TIA)
Uses symbolic variable names
`Conveyor_Start_PB`
`Tank_Level_Scaled`
Hardware mapped vis I/O aliasing

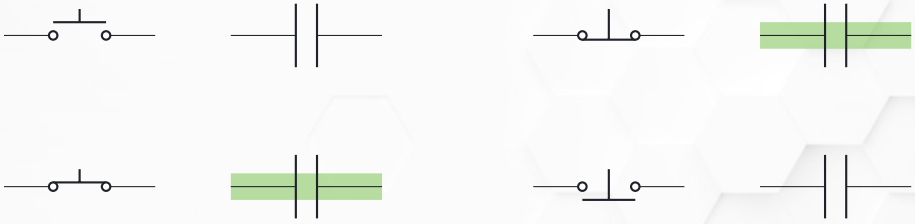
Basic PLC Programming Instructions

The 3 Basic Instructions

- Ladder logic doesn't care what the input *is*
- All that matters is on/off
- Remember, these were intentionally designed to look like relay contacts and coils

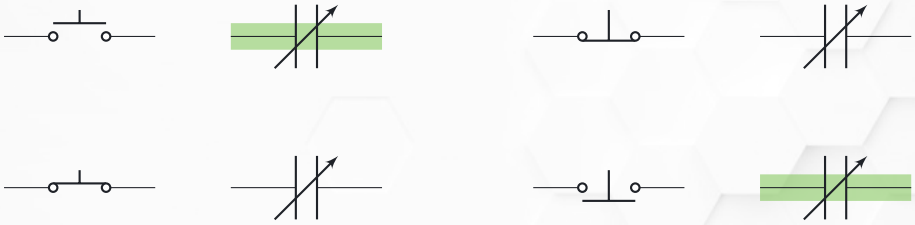
Examine if Closed (XIC)

True when the input is true (on)



Examine if Open (XIO)

True when the input is false (off)

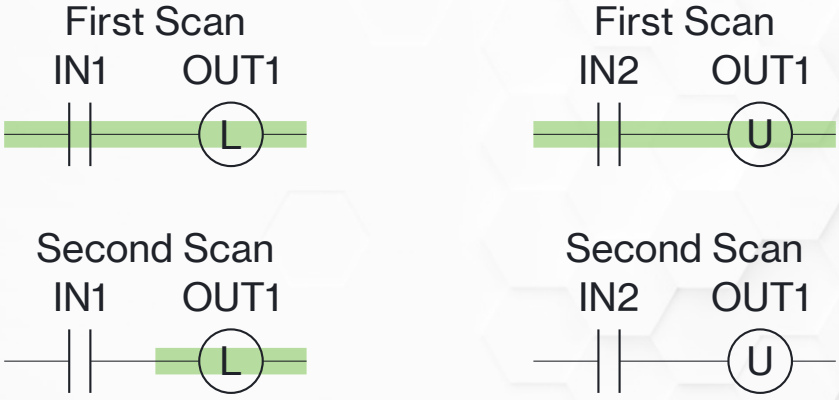


Output Energize (OTE)

True when the logic before it is true

Output Latch (OTL) and Unlatch (OTU)

Once turned on the outputs *stays on* until explcitly unlatched



Take Caution with Latches

- Be *very* careful with latches–my recommendation is to not use them as much as possible
- OTE is *volatile*–when power cycles the output is set to 0 (off)
- OTL is *NOT* volatile–if it's on when power goes out it will be on when power is restored
- This can and has caused unwanted/unsafe machine motion

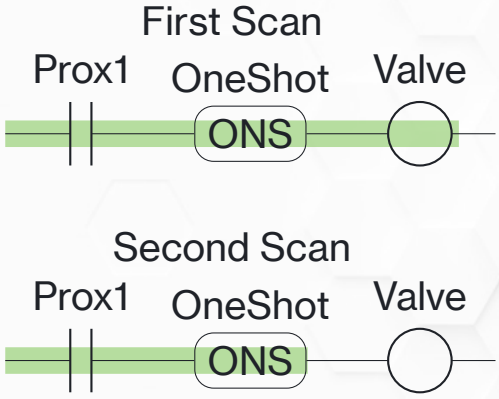
Latch/Unlatch on Key Platforms

How Key Platforms Handle Latch/Unlatch on Power Cycle

Manufacturer	Equivalent Instructions	Power Cycle Behavior	Memory Control Mehcanism
Allen-Bradley (Logix5000)	OTL/OTU	Retains state across power cycles (non-volatile)	Tied directly to the instruction choice (OTL/OTU memory retention vs. OTE non-retention)
Siemens (TIA Portal)	S (Set)/ R (Reset)	Resets to `0` unless declared as Retentive	Bit/DB retain settings in tag table
Beckhoff (TwinCAT)/ CODESYS	S (Set)/ R (Reset)	Resets to `0` unless located in Retain memory	`RETAIN` or `PERSISTENT` keywords
Mitsubishi (GX Works)	SET/RST	Retains state if mapped to Latch Range	Latch (L) device memory range
Omron (Sysmac Studio)	Set/Rst	Resets to `0` unless declared as Retentive	Retain attribute checkbox on VAR

Oneshot

A oneshot is true for *one scan* while the logic before it is true–the logic has to turn false to reset it



PLC Timers and Counters

Timers and Counters

- Timers time, counters count–but their under-the-hood structure is *very* similar
- “Timer” and “counter” refer to the *variable type*
- The timer or counter variable is then used in conjunction with a timer or counter instruction

Timer and Counter Variable Types

- Timers and counters are made up of 3 words:
  1. Word 0: Control word–used as bits, what the bits *do* vary between timers and counters
  2. Word 1: Preset (PRE)–the time/count target
  3. Word 2: Accumulated (ACC)–the current time/count

Timer Variables

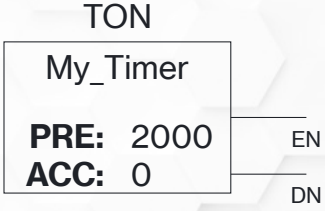
EN Rung logic before the instruction is true
TT The timer is actively timing
DN The preset value has been reached
PRE The target time, in *milliseconds*
ACC The current timer value, in *milliseconds*

Timer Variable Warning

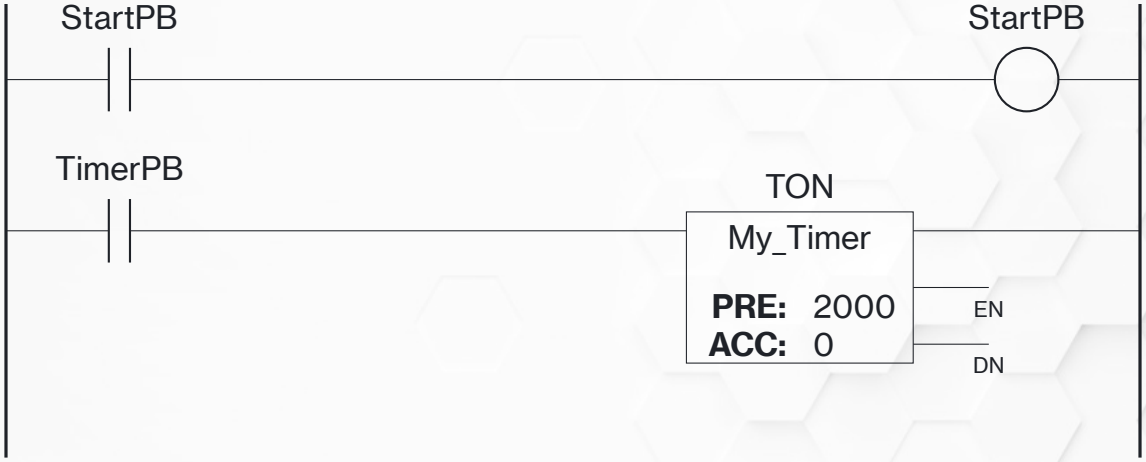
- Timer variables can only be used *once*
- Software will *allow* you to use them more than once, but you will definitely see fun and interesting behavior
- One timer variable per timer instruction

Timer On Instruction

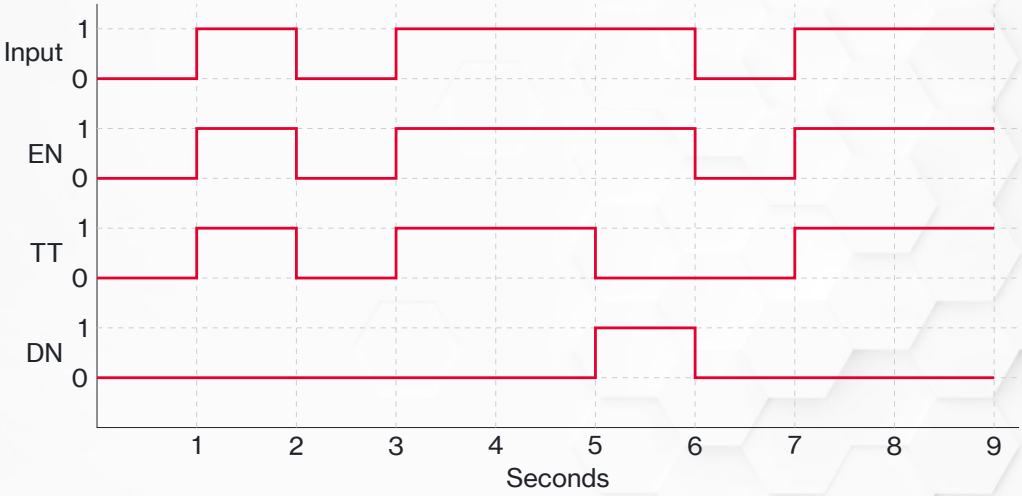
- Starts timing when the preceding logic is true
- Stops timing when the logic goes false
- `ACC` is cleared when the logic goes false



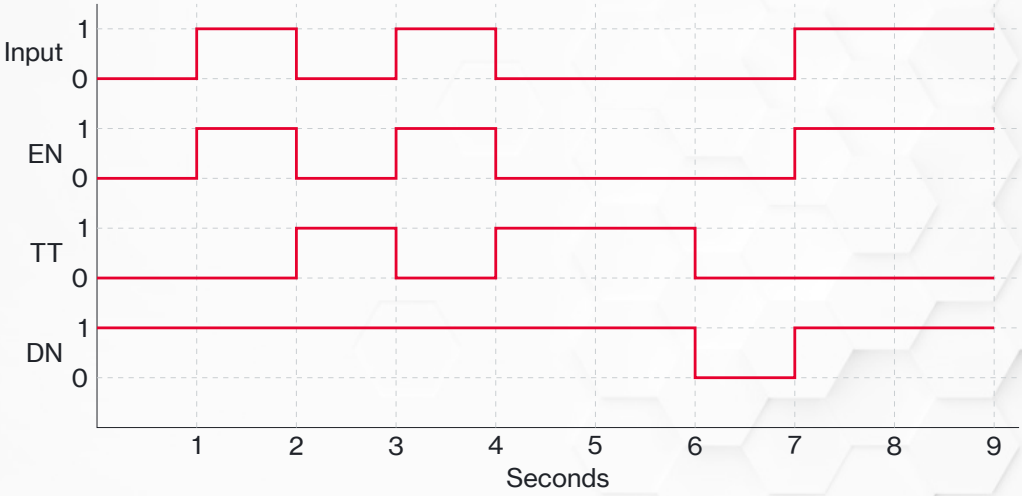
TON Example



TON Timing Diagram



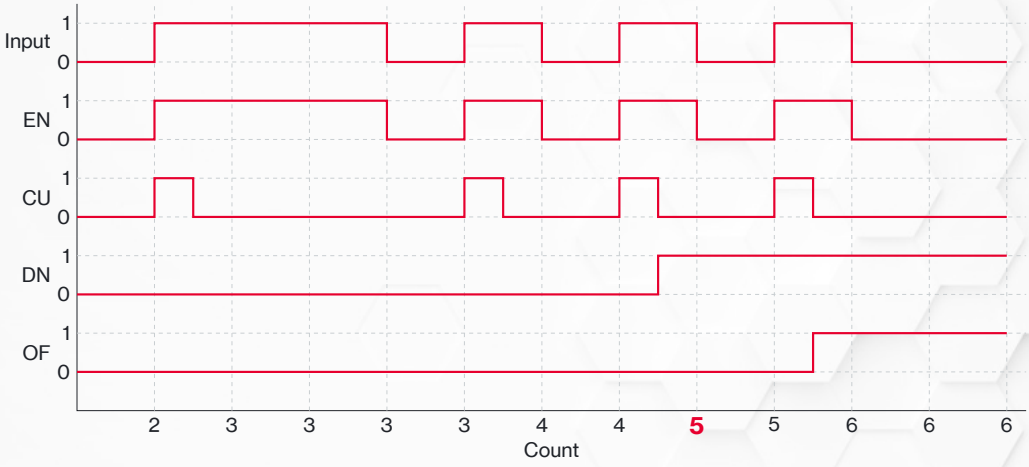
TOF Timing Diagram



Counter Variables

CU The logic before the `CTU` instruction is true
CD The logic before the `CTD` instruction is true
DN The preset value has been reached
OV The `ACC` has exceeded the DINT limit of 2,147,483,647
UN The `ACC` has exceed the DINT limit of -2,147,483,648
PRE The target count
ACC The current count

CTU Timing Diagram



Advanced PLC Programming Instructions

Data Manupulation Instructions

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